SHARP LOG-SOBOLEV INEQUALITIES ON FINITE CYCLIC GROUPS

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Abstract. Let $\mathbb{Z}_n$ be the cyclic group equipped with the uniform probability measure π, and let A_{ψ_n} be the Laplacian with word length

$$\psi_n(k) = \min(k, n-k).$$

We prove the sharp log-Sobolev inequality

$$\mathrm{Ent}_\pi(f^2) \le 2\pi(fA_{\psi_n}f), \qquad f : \mathbb{Z}_n \to [0,\infty),$$

for every $n \ge 4$. The proof is inspired by the recent work of Frank and Ivanisvili [FI26] on a sharp log-Sobolev inequality for nearest-neighbor simple random walk. We use their cubic-majorant reduction, which turns the problem into a 3rd moment estimate; the new point is a blockwise 3rd moment estimate adapted to the word-length multiplier. The same 3rd moment argument also recovers the log-Sobolev inequality for Poisson-semigroup on the circle, first proved by Weissler [Wei80]. The same sharp inequalities were also obtained recently by Yao [Yao26] by a different method.

Key words: Log-Sobolev inequality, hypercontractivity, finite cyclic groups, Poisson semigroups, cubic Sobolev inequalities.

1. Introduction

There are two natural translation-invariant dynamics on the discrete circle, or cyclic group, $\mathbb{Z}_n$. The first is the nearest-neighbor simple random walk: from a point, one jumps to either neighbor with equal probability. The associated positive Laplacian is

$$L_n(f)(k) = f(k) - \frac{1}{2}(f(k+1) + f(k-1)).$$

Using the Fourier expansion

$$f = \sum_{k\in\mathbb{Z}_n} \widehat{f}(k)\chi_k, \qquad \chi_k(x) := e^{2\pi \mathrm{i} kx/n},$$

we may write L_n as a Fourier multiplier

$$L_n(f) = \sum_{k\in\mathbb{Z}_n} r_n(k)\widehat{f}(k)\chi_k, \qquad r_n(k) := 1 - \cos(2\pi k/n).$$

The second can be described by the positive word-length Laplacian A_{ψ_n}, which is the Fourier multiplier

$$A_{\psi_n} f = \sum_{k\in\mathbb{Z}_n} \psi_n(k)\widehat{f}(k)\chi_k, \qquad \psi_n(k) := \min(k, n-k).$$

Equivalently, $-A_{\psi_n}$ is the generator of the Poisson semigroup. Here, one can think about ψ_n as the word length function with respect to the symmetric generator set $\{-1, 1\}$.

In this paper, we shall consider sharp log-Sobolev inequalities, or equivalently hypercontractivity, in these two cases. For classical log-Sobolev inequalities and hypercontractivity, we refer to [Bon70, Gro75, Bec75, Gro06] and references therein for background.

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We shall use the convention that π denotes the uniform probability measure on $\mathbb{Z}_n$. Thus, for all f on $\mathbb{Z}_n$,

$$\pi(f) = \frac{1}{n} \sum_{x \in \mathbb{Z}_n} f(x).$$

For all non-negative f on $\mathbb{Z}_n$, we define

$$\mathrm{Ent}_\pi(f) := \pi(f \log f) - \pi(f) \log \pi(f)\,.$$

For the nearest-neighbor cycle, the log-Sobolev inequality states that for all $f : \mathbb{Z}_n \to \mathbb{R}$

$$\mathrm{Ent}_\pi\big(f^2\big) \le c\pi(f L_n f)\,. \tag{1.1}$$

The problem of determining the sharp log-Sobolev constant c goes back to the work of Diaconis and Saloff-Coste on finite Markov chains [DS96]. After partial progress by Chen and Sheu [CS03], Chen, Liu and Saloff-Coste [CLS08], and Faust and Fawzi [FF24], Frank and Ivanisvili settled the problem for all $n \ge 4$, proving that the optimal constant is $c = 2/r_n(1)$ [FI26]. Their key new step is an optimal cubic Sobolev inequality, which inspired this work.

For the word-length Laplacian, the log-Sobolev inequality reads: for all $f : \mathbb{Z}_n \to \mathbb{R}$

$$\mathrm{Ent}_\pi\big(f^2\big) \le c\pi(f A_{\psi_n} f)\,. \tag{1.2}$$

The study of the optimal constant c appeared in works of Beckner, Janson and Jerison [BJJ83], Andersson [And02], Wolff [Wol07], Junge, Palazuelos, Parcet and Perrin [JPPP17]. In two preprints, Yao first [Yao25] settled the special cases $n = 2^r$ and $n = 3 \cdot 2^r$, and then, more recently, all $n \ge 4$ [Yao26].

We remark that the log-Sobolev inequality for the word-length (1.2) is stronger than that of the nearest-neighbor simple random walk (1.1). Indeed, both generators share the same eigenvectors, the characters χ_k's. Their eigenvalues, after normalization so that the spectral gap is one, have the following relation:

$$\ell_n(k) := \frac{r_n(k)}{r_n(1)} \ge \psi_n(k), \qquad 1 \le k \le n-1. \tag{1.3}$$

To see this, write $m = \min\{k, n-k\} \le n/2$, we have

$$\frac{r_n(k)}{r_n(1)} = \frac{1-\cos(2\pi k/n)}{1-\cos(2\pi/n)} = \frac{1-\cos(m2\pi/n)}{1-\cos(2\pi/n)} \ge m = \psi_n(k), \qquad 1 \le k \le n-1, \tag{1.4}$$

where the inequality is by induction and the identity

$$1-\cos(a+b)-(1-\cos a)-(1-\cos b) = 4\cos\left(\frac{a+b}{2}\right)\sin\left(\frac{a}{2}\right)\sin\left(\frac{b}{2}\right) \ge 0, \qquad a,b \ge 0,\ a+b \le \pi.$$

The main result of this paper is the proof of the sharp log-Sobolev inequality for the word-length case for all $n \ge 4$. The proof is inspired by Frank–Ivanisvili and is different from Yao's method.

Theorem 1. *For every $n \ge 4$ and every nonnegative $f : \mathbb{Z}_n \to \mathbb{R}$,*

$$\mathrm{Ent}_\pi\big(f^2\big) \le 2\pi(f A_{\psi_n} f)\,.$$

The constant 2 is sharp.

For complex-valued functions, the corresponding form is

$$\mathrm{Ent}_\pi\big(|f|^2\big) \le 2\pi\big(\overline{f}\, A_{\psi_n} f\big)\,,$$

and it follows from the real nonnegative statement by the standard contraction of Dirichlet forms. Following standard arguments, one obtains the following hypercontractivity result of the Poisson semigroup $P_t = e^{-tA_{\psi_n}}$. Sufficiency follows from Gross's theorem [Gro75] applied to Theorem 1; necessity follows from the standard perturbation.

Corollary 2 (Hypercontractivity)**.** *Let $P_t = e^{-tA_{\psi_n}}$. For $1 < p \le q < \infty$,*

$$\|P_t f\|_q \le \|f\|_p \qquad \text{for all } f : \mathbb{Z}_n \to \mathbb{C}$$

holds if and only if

$$t \ge \frac{1}{2} \log \frac{q-1}{p-1}.$$

Remark 3. It is a standard argument that the above log-Sobolev inequality and hypercontractivity extend to products of $\mathbb{Z}_n$ with the same constants.

Our proof of Theorem 1 uses the cubic-majorant reduction idea of Frank–Ivanisvili who observed that

$$2t^2 \log t \le \frac{2}{3}(t-1)^2(t+2) + (t^2-1), \qquad t \ge 0.$$

See Lemma 5 below. This reduces the problem to a cubic, or 3rd moment, Sobolev inequality. However, the final high-frequency estimate in their nearest-neighbor proof does not adapt directly to the word-length multiplier. In fact, the main obstacle to a literal adaptation of [FI26] can be seen from the proof of their [FI26, Lemma 3], which uses

$$\sum_{j=2}^{n-2} \frac{1}{\ell_n(j) - 1} < \frac{3}{2}.$$

However, for the word-length multiplier, $\psi_n(j) - 1 = j - 1$ when $j < n/2$, so the analogous estimate would contain

$$\sum_{j=2}^{m} \frac{1}{j-1} \simeq \log m,$$

which produces a logarithmic loss.

Our replacement for this final step is a blockwise 3rd moment estimate. On the Fourier side, we organize the proof by conjugate pairs of frequencies $\{j, -j\}$, except in the even case where the middle frequency $n/2$ is self-conjugate. If $u : \mathbb{Z}_n \to \mathbb{R}$ has mean zero and

$$S^2 = \pi\big(u^2\big), \qquad Q^2 = \pi(uA_{\psi_n} u) - S^2, \qquad S, Q \ge 0, \tag{1.5}$$

then we prove

$$\big|\pi\big(u^3\big)\big| \le 3S^2 Q.$$

This 3rd moment estimate counts only the cubic Fourier resonances that survive averaging. For odd cyclic groups these resonances are

$$i + j = k, \qquad i + j + k = n.$$

For even cyclic groups the same resonances occur, and one must also handle the self-conjugate middle frequency $n/2$. Compared with Frank–Ivanisvili, the main addition is this blockwise 3rd moment estimate: it closes the cubic inequality for the linear word-length spectrum without summing the harmonic reciprocal-surplus weights.

We close the introduction with the continuous-circle counterpart. Let $\mathbb{T} = \mathbb{R}/\mathbb{Z}$, let π be normalized Haar measure, and let $A_{\mathbb{T}}$ be the Poisson generator on $\mathbb{T}$,

$$A_{\mathbb{T}}(f) = \sum_{k \in \mathbb{Z}} |k| \ \widehat{f}(k) \chi_k, \qquad \chi_k(x) = e^{2\pi \mathrm{i} k x},$$

so that

$$\int_{\mathbb{T}} f A_{\mathbb{T}} f \, d\pi = \sum_{k \in \mathbb{Z}} |k| \ \left|\widehat{f}(k)\right|^2.$$

Weissler [Wei80] first proved the sharp Poisson-semigroup log-Sobolev inequality on the circle.

Theorem 4. *For every nonnegative trigonometric polynomial f on $\mathbb{T}$,*

$$\mathrm{Ent}_\pi\big(f^2\big) \le 2\int_{\mathbb{T}} f A_{\mathbb{T}} f \, d\pi.$$

This log-Sobolev inequality for Poisson semigroup can be recovered from the finite word-length inequalities by a Riemann-sum limit; see [Yao26]. Frank and Ivanisvili used the analogous limiting argument for the nearest-neighbor multiplier, which yields the log-Sobolev inequality for the heat-semigroup on $\mathbb{T}$ [FI26]. Indeed, after spectral-gap normalization, the nearest-neighbor multiplier

$$\ell_n(j) = \frac{1-\cos(2\pi j/n)}{1-\cos(2\pi/n)}$$

satisfies $\ell_n(k) \to k^2$ for each fixed $k > 0$, so its limiting Dirichlet form is $\sum_{k\in\mathbb{Z}} k^2 \left|\widehat{f}(k)\right|^2$, corresponding to the heat semigroup. By contrast, the word-length multiplier $\psi_n(k) = \min\{k, n-k\}$ satisfies $\psi_n(k) \to k$. Thus the word-length limit yields the Poisson Dirichlet form $\sum_{k\in\mathbb{Z}} |k| \left|\widehat{f}(k)\right|^2$.

Similar to the finite cyclic-group proof, the direct 3rd moment argument also extends to the circle group case without any difficulty. In Section 5 we provide the details.

2. The cubic reduction

For any $f : \mathbb{Z}_n \to \mathbb{R}$, set

$$\mathcal{E}_n(f) := \pi(f A_{\psi_n} f) = \sum_{k=0}^{n-1} \psi_n(k) \left|\widehat{f}(k)\right|^2.$$

We recall the cubic majorant used in Frank and Ivanisvili's cubic Sobolev reduction [FI26].

Lemma 5 (Cubic majorant [FI26])**.** *For every $t \ge 0$,*

$$2t^2 \log t \le \frac{2}{3}(t-1)^2(t+2) + (t^2-1),$$

where the left side is interpreted as 0 at $t = 0$.

Proposition 6. *Suppose that for every $x : \mathbb{Z}_n \to [0,\infty)$ with $\pi\big(x^2\big) = 1$,*

$$\mathcal{E}_n(x) \ge \frac{1}{3}\pi\big((x-1)^2(x+2)\big). \tag{2.1}$$

Then

$$\mathrm{Ent}_\pi\big(f^2\big) \le 2\mathcal{E}_n(f)$$

for every nonnegative $f : \mathbb{Z}_n \to \mathbb{R}$.

Proof. By homogeneity, normalize $\pi\big(f^2\big) = 1$. Then

$$\mathrm{Ent}_\pi\big(f^2\big) = 2\pi\big(f^2 \log f\big).$$

Applying Lemma 5 pointwise and averaging gives

$$\mathrm{Ent}_\pi\big(f^2\big) \le \frac{2}{3}\pi\big((f-1)^2(f+2)\big) + \pi\big(f^2-1\big) = \frac{2}{3}\pi\big((f-1)^2(f+2)\big).$$

The conclusion follows from (2.1). □

The next proposition isolates the estimate needed for the cubic criterion (2.1).

Proposition 7. *Fix $n \ge 4$. Assume that every real mean-zero $u : \mathbb{Z}_n \to \mathbb{R}$ satisfies*

$$\big|\pi\big(u^3\big)\big| \le 3S^2Q, \qquad S^2 = \pi\big(u^2\big), \quad Q^2 = \mathcal{E}_n(u) - S^2, \quad S, Q \ge 0. \tag{2.2}$$

Then (2.1) holds on $\mathbb{Z}_n$.

Proof. Let $x : \mathbb{Z}_n \to [0,\infty)$ and normalize $\pi\big(x^2\big) = 1$. Put

$$a = \pi(x)\,, \qquad u = x - a.$$

Then $\pi(u) = 0$, $0 \le a \le 1$, and

$$S^2 = \pi\big(u^2\big) = 1 - a^2, \qquad \mathcal{E}_n(x) = \mathcal{E}_n(u) = S^2 + Q^2.$$

Since $(t-1)^2(t+2) = t^3 - 3t + 2$,

$$\pi\big((x-1)^2(x+2)\big) = \pi\big(x^3\big) - 3a + 2.$$

Expanding $x = a + u$ gives

$$\pi\big(x^3\big) = a^3 + 3aS^2 + \pi\big(u^3\big) = a^3 + 3a(1-a^2) + \pi\big(u^3\big)\,.$$

Hence

$$\pi\big((x-1)^2(x+2)\big) = 2 - 2a^3 + \pi\big(u^3\big)\,. \tag{2.3}$$

Therefore (2.1) is equivalent to

$$\pi\big(u^3\big) \le (1-a)^2(1+2a) + 3Q^2. \tag{2.4}$$

By (2.2),

$$\pi\big(u^3\big) \le 3S^2Q.$$

Also

$$3S^2Q \le \frac{3}{4}S^4 + 3Q^2,$$

and, since $S^2 = 1 - a^2$,

$$(1-a)^2(1+2a) - \frac{3}{4}S^4 = (1-a)^2(1+2a) - \frac{3}{4}(1-a^2)^2 = \frac{1}{4}(1-a)^3(1+3a) \ge 0.$$

Thus (2.4) holds. □

3. The 3rd moment estimate for odd cyclic groups

Assume $n = 2m+1 \ge 5$. For a real mean-zero function $u : \mathbb{Z}_n \to \mathbb{R}$, write

$$u = \sum_{j=1}^{m} \left(c_j\chi_j + \overline{c_j}\chi_{-j}\right),$$

where $\chi_{-j} = \chi_{n-j}$. Define

$$a_j = \sqrt{2}\,|c_j|\,, \qquad 1 \le j \le m.$$

Then

$$S^2 = \pi\big(u^2\big) = \sum_{j=1}^{m} a_j^2, \qquad Q^2 = \sum_{j=2}^{m}(j-1)a_j^2, \qquad S, Q \ge 0.$$

Proposition 8 (Odd 3rd moment estimate)**.** *For every real mean-zero $u : \mathbb{Z}_{2m+1} \to \mathbb{R}$, $m \ge 2$,*

$$\left|\pi\big(u^3\big)\right| \le 3S^2Q.$$

Proof. If $Q = 0$, then u is supported on the first frequency block. Since $n \ge 5$ is odd, no three elements of $\{1,-1\}$ sum to zero modulo n. Hence $\pi\big(u^3\big) = 0$. Assume $Q > 0$.

The zero-frequency cubic resonances are precisely

$$i + j = k, \qquad i + j + k = n, \qquad 1 \le i, j, k \le m,$$

together with conjugate sign patterns. For $2 \le k \le m$, put

$$L_k = \sum_{\substack{i+j=k \\ 1\le i,j\le m}} a_i a_j, \qquad L_1 = 0,$$

and for $1 \le k \le m$, put

$$R_k = \sum_{\substack{i+j=n-k \\ 1\le i,j\le m}} a_i a_j.$$

Coefficient counting gives

$$\left|\pi\big(u^3\big)\right| \le \frac{1}{\sqrt{2}} \sum_{k=1}^{m} a_k(3L_k + R_k). \tag{3.1}$$

Indeed, a product corresponding to $i + j = k$ has magnitude $a_i a_j a_k/(2\sqrt{2})$; the three placements of the negative frequency and the conjugate sign pattern give the contribution $3a_k L_k/\sqrt{2}$. For $i + j + k = n$, the ordered all-positive triples and the all-negative conjugates give $a_k R_k/\sqrt{2}$.

The boundary term $k = 1$ is not covered by the Q^2-weight. Since R_1 corresponds to $i + j = 2m$, the only ordered pair is (m, m), and $R_1 = a_m^2$. Thus

$$B_0 = \frac{1}{\sqrt{2}} a_1 R_1 = \frac{1}{\sqrt{2}} a_1 a_m^2.$$

Using

$$S^2 \ge a_1^2 + a_m^2 \ge 2a_1 a_m, \qquad Q^2 \ge (m-1)a_m^2,$$

we get

$$B_0 \le \frac{1}{2\sqrt{2(m-1)}} S^2 Q \le \frac{1}{2\sqrt{2}} S^2 Q. \tag{3.2}$$

It remains to estimate

$$B_1 = \frac{1}{\sqrt{2}} \sum_{k=2}^{m} a_k(3L_k + R_k).$$

By Cauchy–Schwarz with weight $k - 1$,

$$B_1 \le \frac{1}{\sqrt{2}} Q \left(\sum_{k=2}^{m} \frac{(3L_k + R_k)^2}{k-1} \right)^{1/2}.$$

We claim that

$$\sum_{k=2}^{m} \frac{(3L_k + R_k)^2}{k-1} \le 11 S^4. \tag{3.3}$$

For $2 \le k \le m$, set

$$U_k = \sum_{\substack{i+j=k \\ 1\le i,j\le m}} a_i^2 a_j^2, \qquad V_k = \sum_{\substack{i+j=n-k \\ 1\le i,j\le m}} a_i^2 a_j^2.$$

The sum L_k has $k - 1$ ordered terms, while R_k has k ordered terms. Hence

$$L_k^2 \le (k-1)U_k, \qquad R_k^2 \le kV_k.$$

Since $k/(k-1) \le 2$,

$$\frac{(3L_k + R_k)^2}{k-1} \le \left(3\sqrt{U_k} + \sqrt{2}\sqrt{V_k}\right)^2 \le 11(U_k + V_k).$$

The U_k terms cover ordered pairs with $2 \le i + j \le m$, and the V_k terms cover ordered pairs with $m + 1 \le i + j \le 2m - 1$. These ranges are disjoint and contained in all ordered pairs $1 \le i, j \le m$. Thus

$$\sum_{k=2}^{m} (U_k + V_k) \le \sum_{i,j=1}^{m} a_i^2 a_j^2 = S^4,$$

which proves (3.3). Therefore

$$B_1 \le \sqrt{\frac{11}{2}} S^2 Q. \tag{3.4}$$

Combining (3.1), (3.2), and (3.4),

$$\left|\pi(u^3)\right| \le \left(\sqrt{\frac{11}{2}} + \frac{1}{2\sqrt{2}}\right) S^2 Q < 3S^2 Q.$$

□

4. The 3rd moment estimate for even cyclic groups

Let $n = 2m \ge 4$. The proof is parallel to the odd case, with one change: the middle frequency $m = n/2$ is self-conjugate. Write a real mean-zero function as

$$u = \sum_{j=1}^{m-1} (c_j \chi_j + \overline{c_j} \chi_{-j}) + b\chi_m, \qquad b \in \mathbb{R},$$

where $\chi_m(x) = (-1)^x$. Again, the paired frequencies are only conjugate pairs coming from real-valuedness. The middle frequency is its own negative modulo $2m$, so its coefficient b is real. Put

$$a_j = \sqrt{2}\,|c_j| \quad (1 \le j \le m-1), \qquad a_m = |b|\,.$$

Then

$$S^2 = \sum_{j=1}^{m} a_j^2, \qquad Q^2 = \sum_{j=2}^{m} (j-1) a_j^2, \qquad S, Q \ge 0.$$

Proposition 9 (Even 3rd moment estimate)**.** *For every real mean-zero $u : \mathbb{Z}_{2m} \to \mathbb{R}$, $m \ge 2$,*

$$\left|\pi(u^3)\right| \le 3S^2 Q.$$

Proof. The case $Q = 0$ is again trivial, because then only the first frequency block is present and no cubic zero-frequency resonance occurs for $n \ge 4$. Assume $Q > 0$.

For $2 \le k \le m-1$, define

$$L_k = \sum_{\substack{i+j=k \\ 1 \le i,j \le m-1}} a_i a_j, \qquad R_k = \sum_{\substack{i+j=2m-k \\ 1 \le i,j \le m-1}} a_i a_j,$$

and define the middle-frequency sum

$$L_m = \sum_{\substack{i+j=m \\ 1 \le i,j \le m-1}} a_i a_j.$$

Empty sums are understood as zero. The ordinary and wrap-around resonances give the same coefficients as in the odd case, while the self-conjugate middle frequency contributes with coefficient 3. Hence

$$\left|\pi(u^3)\right| \le \frac{1}{\sqrt{2}} \sum_{k=2}^{m-1} a_k (3L_k + R_k) + 3a_m L_m. \tag{4.1}$$

By Cauchy–Schwarz, one has

$$\left|\pi(u^3)\right| \le Q\sqrt{W},$$

where

$$W = \sum_{k=2}^{m-1} \frac{(3L_k + R_k)^2}{2(k-1)} + \frac{9L_m^2}{m-1}.$$

We prove

$$W \le 9S^4. \tag{4.2}$$

For $2 \le k \le m-1$, let

$$U_k = \sum_{\substack{i+j=k \\ 1\le i,j\le m-1}} a_i^2 a_j^2, \qquad V_k = \sum_{\substack{i+j=2m-k \\ 1\le i,j\le m-1}} a_i^2 a_j^2.$$

Both L_k and R_k have $k-1$ ordered terms, so

$$L_k^2 \le (k-1)U_k, \qquad R_k^2 \le (k-1)V_k.$$

Therefore

$$\frac{(3L_k + R_k)^2}{2(k-1)} \le \frac{1}{2}\left(3\sqrt{U_k} + \sqrt{V_k}\right)^2 \le 5(U_k + V_k),$$

using $(3a+b)^2 \le 10(a^2+b^2)$. For the middle-frequency term, put

$$U_m = \sum_{\substack{i+j=m \\ 1\le i,j\le m-1}} a_i^2 a_j^2.$$

Since L_m has $m-1$ ordered terms,

$$\frac{9L_m^2}{m-1} \le 9U_m.$$

The U_k terms cover sums $2 \le i+j \le m-1$, the U_m term covers sum m, and the V_k terms cover sums $m+1 \le i+j \le 2m-2$. These families partition the ordered pairs $1 \le i,j \le m-1$. Hence

$$\sum_{k=2}^{m-1}(U_k + V_k) + U_m = \left(\sum_{j=1}^{m-1} a_j^2\right)^2.$$

Consequently

$$W \le 5\left(\sum_{j=1}^{m-1} a_j^2\right)^2 + 4U_m \le 9\left(\sum_{j=1}^{m-1} a_j^2\right)^2 \le 9S^4.$$

This proves (4.2), and therefore

$$\left|\pi\left(u^3\right)\right| \le Q\sqrt{9S^4} = 3S^2Q.$$

□

5. Proofs of the main results

Proof of Theorem 1. If $n \ge 4$ is odd, then $n = 2m+1$ with $m \ge 2$, and Proposition 8 verifies the hypothesis of Proposition 7. If $n \ge 4$ is even, then Proposition 9 verifies the same hypothesis. Thus the cubic criterion holds for all $n \ge 4$, and Proposition 6 gives

$$\mathrm{Ent}_\pi\left(f^2\right) \le 2\pi(fA_{\psi_n}f)\,.$$

The sharpness of the constant is a standard perturbation argument, since the spectral gap is 1. □

We finish with a continuous-circle proof for completeness.

Proof of Theorem 4. For completeness, we also record the direct 3rd moment estimate. As in Proposition 7, it is enough to prove the circle analogue of (2.2). Let u be a real mean-zero trigonometric polynomial and write

$$u = \sum_{j\ge 1}\left(c_j\chi_j + \overline{c_j}\chi_{-j}\right), \qquad a_j = \sqrt{2}\,|c_j|\,.$$

Only finitely many of the a_j's are nonzero. Then

$$S^2 = \int_{\mathbb{T}} u^2 \, d\pi = \sum_{j\geq 1} a_j^2, \qquad Q^2 = \int_{\mathbb{T}} u A_{\mathbb{T}} u \, d\pi - S^2 = \sum_{j\geq 2} (j-1) a_j^2, \qquad S, Q \geq 0.$$

If $Q = 0$, only the first frequency block is present and no cubic zero-frequency resonance occurs, so $\int_{\mathbb{T}} u^3 \, d\pi = 0$. Assume $Q > 0$. The only zero-frequency cubic resonances are $i + j = k$, together with their conjugates. For $k \geq 2$, put

$$L_k = \sum_{i+j=k} a_i a_j, \qquad U_k = \sum_{i+j=k} a_i^2 a_j^2, \qquad i, j \geq 1.$$

Coefficient counting gives

$$\left| \int_{\mathbb{T}} u^3 \, d\pi \right| \leq \frac{3}{\sqrt{2}} \sum_{k\geq 2} a_k L_k.$$

By Cauchy–Schwarz, we get

$$\left| \int_{\mathbb{T}} u^3 \, d\pi \right| \leq \frac{3}{\sqrt{2}} Q \left(\sum_{k\geq 2} \frac{L_k^2}{k-1} \right)^{1/2}.$$

Since L_k has $k-1$ ordered terms,

$$L_k^2 \leq (k-1) U_k.$$

Therefore

$$\sum_{k\geq 2} \frac{L_k^2}{k-1} \leq \sum_{k\geq 2} U_k = \sum_{i,j\geq 1} a_i^2 a_j^2 = S^4.$$

Thus

$$\left| \int_{\mathbb{T}} u^3 \, d\pi \right| \leq \frac{3}{\sqrt{2}} S^2 Q \leq 3 S^2 Q.$$

The cubic reduction gives $\mathrm{Ent}_\pi(f^2) \leq 2 \int_{\mathbb{T}} f A_{\mathbb{T}} f \, d\pi$ for nonnegative trigonometric polynomials f, and this concludes the proof. □

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